ON THE STATE SPACE OF A LINEAR DIGITAL FILTER

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Abstract. The time-variant behaviour of a linear discrete system is studied by means of the state space representation. Proceeding from an observation on the flow of energy inside the capacitor of an analogue filter, we bring first- and second-order systems into optimised time-variant and time-invariant recursive stable structures. The transformation of a continuous state space into its discrete counterpart is carried out, with the equilibrium matrix and the eigenspaces of the transition matrix playing the key role.

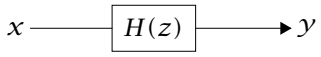
1. INTRODUCTION

Once the transfer function of a linear digital system has been obtained, the question of how to realise it efficiently in real time presents itself. For recursive filters, this question admits no unique answer: there exist infinitely many structurally distinct realisations, differing in their simplicity of implementation, speed of execution, and robustness with respect to numerical stability. In approaching any problem, it is therefore a matter of principle to begin with the simplest case, extract from it all available information, and only then to advance to the problem of greater generality. In Section 3, the recursive structure is brought into state space form and a direct correspondence with the analogue RC circuit is established. Stability is taken up in Section 4, where sufficient conditions are given for both the time-invariant and time-variant cases. Finally, a second-order continuous state space of known stability is discretised by eigendecomposition, and the resulting form is augmented by a free parameter without alteration of the frequency response. Throughout, each structure is reduced to an optimised and elegant update form employing the coefficients of [1], with operation counts and stability properties made explicit at every stage.

2. THE DIRECT FORM

2.1. The transfer function

A linear time-invariant system is characterised entirely by its transfer function $H(z)$, which expresses the direct relation between the input x and the output y.



(2.1)

Thus, given the input vector $\boldsymbol{x}$, one may predict the output vector $\boldsymbol{y}$ by examining $H(z)$, or more precisely, its poles and zeros. We refer to this property as determinacy; one of the mainstays of the theory of ordinary differential equations. For realtime causal systems, however, the input vector arrives sample by sample, and therefore, a recursive form of the transfer function is required. A simple and intuitive transformation accomplishes this, as shown in Table 2.1, using the low-pass form in [1]†

Table 2.1.
Direct Form I mapping of a first- and second-order transfer function.

	First–order	Second–order
$H(z)$	$\frac{1+\beta}{1+\alpha}\cdot\frac{1+\alpha z^{-1}}{1+\beta z^{-1}}$	$\frac{1+\beta_1+\beta_2}{1+\alpha_1+\alpha_2}\cdot\frac{1+\alpha_1 z^{-1}+\alpha_2 z^{-2}}{1+\beta_1 z^{-1}+\beta_2 z^{-2}}$
$\boldsymbol{y}_n$	$\frac{1+\beta}{1+\alpha}\cdot(\boldsymbol{x}_n+\alpha\boldsymbol{x}_{n-1})-\beta\boldsymbol{y}_{n-1}$	$\frac{1+\beta_1+\beta_2}{1+\alpha_1+\alpha_2}\cdot(\boldsymbol{x}_n+\alpha_1\boldsymbol{x}_{n-1}+\alpha_2\boldsymbol{x}_{n-2})-\beta_1\boldsymbol{y}_{n-1}-\beta_2\boldsymbol{y}_{n-2}$

The pattern is evident. Each z^{-n} acts as a state from the past, also known as a delay element. The literature calls this recursive sequence Direct Form I. Although mathematically correct in an ideal world, it exhibits several weaknesses in the real world. We enumerate them for a system of order m.

First, the number of states is $2m$ for a system of order m, whereas the theoretical minimum is m. Second, the operation count stands at $2m$ additions and $2m+1$ multiplications, a figure which, as we shall show, is not optimal. Third, numerical instability arises for low-cut systems with a very small cutoff frequency. Fourth, and most pertinently for our purposes, instability occurs for $m > 1$ under time-varying coefficients.

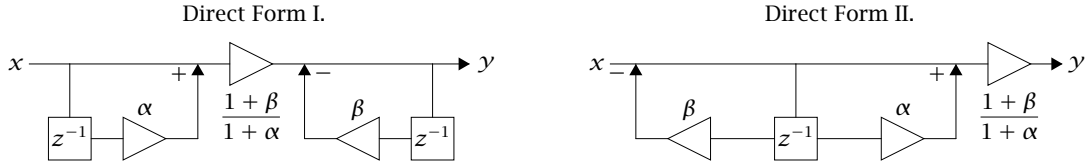


Figure 2.1. Structures directly derived from the transfer function $H(z)$.

By a simple algebraic manipulation, one transforms the Direct Form I into the Direct Form II, as shown in Figure 2.1, thereby minimising the number of states. The third weakness is, however, aggravated by this transformation. It is clear that simple and unsystematic attempts will not suffice; a principled approach to the problem is required.

†The coefficient formulæ of [1] are employed throughout, though the reasoning is not limited to them. Any notation not introduced in the present paper is defined in [1]

3. THE STATE SPACE

3.1. Definition and general form

A linear system of order m in the state space representation has the general form

$$\begin{aligned} \boldsymbol{y}_n &= \boldsymbol{A}_n \boldsymbol{x}_n + \boldsymbol{B}_n \boldsymbol{z}_n \\ \boldsymbol{z}_{n+1} &= \boldsymbol{C}_n \boldsymbol{x}_n + \boldsymbol{D}_n \boldsymbol{z}_n \end{aligned} \tag{3.1}$$

The m-dimensional real vector $\boldsymbol{z}_n$ is composed of a minimum set of states that uniquely describe a system, given the state matrices $\{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}\}_n$, which contain the coefficients of the system and satisfy

$$\boldsymbol{A}_n \in \mathbb{R}^{1\times 1}, \qquad \boldsymbol{B}_n \in \mathbb{R}^{1\times m}, \qquad \boldsymbol{C}_n \in \mathbb{R}^{m\times 1}, \qquad \boldsymbol{D}_n \in \mathbb{R}^{m\times m} \tag{3.2}$$

The transfer function takes the form

$$H_m(z) = \boldsymbol{A} + \boldsymbol{B} \cdot (z - \boldsymbol{D})^{-1} \boldsymbol{C} \tag{3.3}$$

For first- and second-order systems, with $\operatorname{tr} \cdot$ and $\det \cdot$ denoting the trace and determinant of a matrix respectively, this reduces to the more convenient expressions

$$\begin{aligned} H_1(z) &= \boldsymbol{A} \cdot \frac{1 + \left(\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1} - \operatorname{tr} \boldsymbol{D}\right) z^{-1}}{1 - \operatorname{tr} \boldsymbol{D}\, z^{-1}} \\ H_2(z) &= \boldsymbol{A} \cdot \frac{1 + \left(\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1} - \operatorname{tr} \boldsymbol{D}\right) z^{-1} + \left(\boldsymbol{B} \cdot (\boldsymbol{I} - \boldsymbol{D})^{-1} \boldsymbol{C} - \boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1} + \det \boldsymbol{D}\right) z^{-2}}{1 - \operatorname{tr} \boldsymbol{D}\, z^{-1} + \det \boldsymbol{D}\, z^{-2}} \end{aligned} \tag{3.4}$$

Setting aside the pole-independent scalar $G = 1/(1 + \alpha)$ from our first-order transfer function and solving the resulting state space equation, we find

$$H_1(z) = (1 + \beta) \cdot \frac{1 + \alpha z^{-1}}{1 + \beta z^{-1}} \quad \Longrightarrow \quad \begin{cases} \boldsymbol{A} = 1 + \beta \\ \boldsymbol{D} = -\beta \\ \boldsymbol{B}\boldsymbol{C} = (1 + \beta)(\alpha - \beta) \end{cases} \tag{3.5}$$

The set of solutions is infinite. Since the input matrix $\boldsymbol{C}$, unlike the output matrix $\boldsymbol{B}$, is directly related to the evolution of the states, it is important to understand its meaning. Beyond its role in controllability in control theory, it is actually directly linked to the passive elements of an electronic circuit, and it is this link that we now examine.

3.2. The analogue analogy

A first-order analogue filter is composed of a resistance R and a capacitance C. Let us denote by Z the complex impedance as $Z = R - i/2\pi f C$, on which the transfer function $H(z)$ depends. With a cutoff

frequency of $1/2\pi RC$, it is of interest, for time-variant systems, to know whether it is the real or the imaginary part of Z that is varying. In fact, the circuit behaves quite differently depending on the two cases.

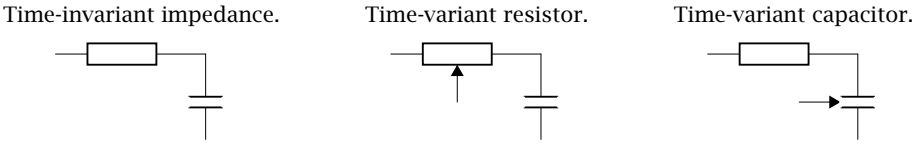


Figure 3.1. Analogue filter with different time behaviours.

The state $\boldsymbol{z}_n$, physically, is comparable to the electric charge Q_t stored on the conductors of a capacitor. This charge converges, as time tends to infinity, to the product of the capacitance C and the voltage U between the conductors. By assigning U to the input x, one may emulate this charging process in the state space. The comparison reads

$$\begin{aligned} Q_t &\xrightarrow{t\to\infty} U \cdot C \\ \boldsymbol{z}_n &\xrightarrow{n\to\infty} U \cdot (\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{C} \end{aligned} \tag{3.6}$$

We observe that a constant capacitance corresponds to a constant matrix $(\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{C}$, whilst a constant resistance corresponds to a constant input matrix $\boldsymbol{C}$. Let us normalise these constants to unity and introduce the stable equilibrium matrix $\boldsymbol{E} \coloneqq (\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{C}$. The state space may then be completed for both time-variant cases.

$$\begin{aligned} \mathfrak{Im}\, Z = 1 \quad &\Longrightarrow \quad \boldsymbol{E} = \boldsymbol{1} \quad \Longrightarrow \quad \begin{cases} \boldsymbol{B} = \alpha - \beta \\ \boldsymbol{C} = 1 + \beta \end{cases} \\ \mathfrak{Re}\, Z = 1 \quad &\Longrightarrow \quad \boldsymbol{C} = \boldsymbol{1} \quad \Longrightarrow \quad \begin{cases} \boldsymbol{B} = (\alpha - \beta)(1 + \beta) \\ \boldsymbol{C} = 1 \end{cases} \end{aligned} \tag{3.7}$$

It turns out that the Direct Form II of Figure 2.1 also simulates a circuit with a time-varying capacitance. Whether this is a desired behaviour is an engineering or artistic preference. In either case, rather than evaluating the state $\boldsymbol{z}_n$ and output $\boldsymbol{y}_n$ directly according to (3.1), we introduce the two auxiliary quantities

$$\vartheta_{\mathfrak{Im}} = (1 + \beta)(\boldsymbol{x}_n - \boldsymbol{z}_n), \qquad \vartheta_{\mathfrak{Re}} = \boldsymbol{x}_n - (1 + \beta)\boldsymbol{z}_n \tag{3.8}$$

and simplify the update accordingly, as recorded in Table 3.1 and Table 3.2.

Table 3.1.
State space update with $\mathfrak{Im}\, Z = 1$.

	$\boldsymbol{y}_n$	$\boldsymbol{z}_{n+1}$
High-pass	$G \cdot \vartheta_{\mathfrak{Im}}$	$\boldsymbol{z}_n + \vartheta_{\mathfrak{Im}}$
Other	$G \cdot \vartheta_{\mathfrak{Im}} + \boldsymbol{z}_n$	$\boldsymbol{z}_n + \vartheta_{\mathfrak{Im}}$

Table 3.2.
State space update with $\mathfrak{Re}\, Z = 1$.

	$\boldsymbol{y}_n$	$\boldsymbol{z}_{n+1}$
High-pass	$(1 + \beta)(G \cdot \vartheta_{\mathfrak{Re}})$	$\boldsymbol{z}_n + \vartheta_{\mathfrak{Re}}$
Other	$(1 + \beta)(G \cdot \vartheta_{\mathfrak{Re}} + \boldsymbol{z}_n)$	$\boldsymbol{z}_n + \vartheta_{\mathfrak{Re}}$

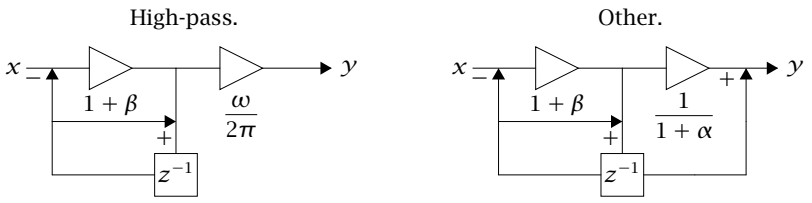


Figure 3.2. Optimised state space update with $\mathfrak{Im}\, Z = 1$.

We remark that the structure corresponding to a fixed capacitance achieves, compared with Direct Form I, a minimum of states, a bound on the norm of the state vector $\boldsymbol{z}_n$ below the norm of the input vector $\boldsymbol{x}_n$, and an exchange of one multiplication for one addition.

3.3. The second-order case

The result of the preceding section admits a natural extension to the second-order system. We now require

$$H_2(z) = (1 + \beta_1 + \beta_2) \cdot \frac{1 + \alpha_1 z^{-1} + \alpha_2 z^{-2}}{1 + \beta_1 z^{-1} + \beta_2 z^{-2}} \tag{3.9}$$

which yields the set of solutions

$$\begin{cases} \boldsymbol{A} = 1 + \beta_1 + \beta_2 \\ \operatorname{tr} \boldsymbol{D} = -\beta_1 \\ \det \boldsymbol{D} = \beta_2 \\ \boldsymbol{BC} = (1 + \beta_1 + \beta_2)(\alpha_1 - \beta_1) \\ \boldsymbol{BE} = (1 + \alpha_1 + \alpha_2) - (1 + \beta_1 + \beta_2) \end{cases} \tag{3.10}$$

Here we have four degrees of freedom where the first-order problem had only one. Nevertheless, we seek to find a solution by insisting on simplicity. Let us restrict the entries of the state matrices to the additive group $(\{1, \alpha_1, \alpha_2, \beta_1, \beta_2\}, +)$, so that only additions and subtractions appear as arithmetic operators. This forces the equilibrium matrix $\boldsymbol{E}$ to take values in the set $\{0, 1\}$, excluding the null matrix. With order $m = 2$, it yields $2^m - 1 = 3$ unique permutations:

$$\boldsymbol{E}_1 = \begin{Vmatrix} 1 \\ 0 \end{Vmatrix}, \quad \boldsymbol{E}_2 = \begin{Vmatrix} 0 \\ 1 \end{Vmatrix}, \quad \boldsymbol{E}_3 = \begin{Vmatrix} 1 \\ 1 \end{Vmatrix} \tag{3.11}$$

The input matrix $\boldsymbol{C}$ is correspondingly restricted, yielding three combination pairs

$$\begin{aligned} \boldsymbol{C}_1 &= \begin{Vmatrix} 0 \\ 1 + \beta_1 + \beta_2 \end{Vmatrix} \equiv \boldsymbol{E}_1 \\ \boldsymbol{C}_2 &= \begin{Vmatrix} 1 + \beta_1 + \beta_2 \\ 1 + \beta_1 + \beta_2 \end{Vmatrix} \equiv \boldsymbol{E}_2 \\ \boldsymbol{C}_3 &= \begin{Vmatrix} 1 + \beta_1 + \beta_2 \\ 0 \end{Vmatrix} \equiv \boldsymbol{E}_3 \end{aligned} \tag{3.12}$$

Solving the state space in each case and discarding all symmetric forms, three complete systems emerge, collected in Table 3.3.

Table 3.3.
Different second-order state space systems.

	$\boldsymbol{A}$	$\boldsymbol{B}$	$\boldsymbol{C}$	$\boldsymbol{D}$
System 1	$\begin{Vmatrix} 1+\beta_1+\beta_2 \end{Vmatrix}$	$\begin{Vmatrix} \alpha_1+\alpha_2-\beta_1-\beta_2 \\ \alpha_1-\beta_1 \end{Vmatrix}^{\mathsf{T}}$	$\begin{Vmatrix} 0 \\ 1+\beta_1+\beta_2 \end{Vmatrix}$	$\begin{Vmatrix} 1 & 1 \\ -1-\beta_1-\beta_2 & -1-\beta_1 \end{Vmatrix}$
System 2	$\begin{Vmatrix} 1+\beta_1+\beta_2 \end{Vmatrix}$	$\begin{Vmatrix} \beta_2-\alpha_2 \\ \alpha_1+\alpha_2-\beta_1-\beta_2 \end{Vmatrix}^{\mathsf{T}}$	$\begin{Vmatrix} 1+\beta_1+\beta_2 \\ 1+\beta_1+\beta_2 \end{Vmatrix}$	$\begin{Vmatrix} \beta_2 & -1-\beta_1-\beta_2 \\ \beta_2 & -\beta_1-\beta_2 \end{Vmatrix}$
System 3	$\begin{Vmatrix} 1+\beta_1+\beta_2 \end{Vmatrix}$	$\begin{Vmatrix} \alpha_1-\beta_1 \\ \alpha_2-\beta_2 \end{Vmatrix}^{\mathsf{T}}$	$\begin{Vmatrix} 1+\beta_1+\beta_2 \\ 0 \end{Vmatrix}$	$\begin{Vmatrix} -\beta_1 & -\beta_2 \\ 1 & 0 \end{Vmatrix}$

One observes that System 3 shares the same transition matrix $\boldsymbol{D}$ as the second-order Direct Form II, differing from it in having constant capacitance rather than constant resistance. Furthermore, seemingly, System 2 is an extension of our optimised first-order structure. Indeed, with the auxiliary quantity

$$\vartheta = (1+\beta_1+\beta_2)(\boldsymbol{x}_n - \boldsymbol{z}_{1|n}) + \beta_2 \boldsymbol{z}_{2|n} \tag{3.13}$$

the state space update simplifies to the form shown in Table 3.4:

Table 3.4.
State space update for System 2.

	$\boldsymbol{y}_{n+1}$	$\boldsymbol{z}_{1\|n+1}$	$\boldsymbol{z}_{2\|n+1}$
Low Cut	$G \cdot (\vartheta - \alpha_2 \boldsymbol{z}_{2\|n})$	$\boldsymbol{z}_{1\|n} + \vartheta$	ϑ
Other	$G \cdot (\vartheta - \alpha_2 \boldsymbol{z}_{2\|n}) + \boldsymbol{z}_{1\|n}$	$\boldsymbol{z}_{1\|n} + \vartheta$	ϑ

We employ the term low-cut to refer to any system whose discrete transfer function has at least one zero at $z = 1$; that is, any high-pass or band-pass filter. The comparison with Figure 3.2 reveals the same natural symmetry; both structures require $2m + 1$ additions and $2m$ multiplications.

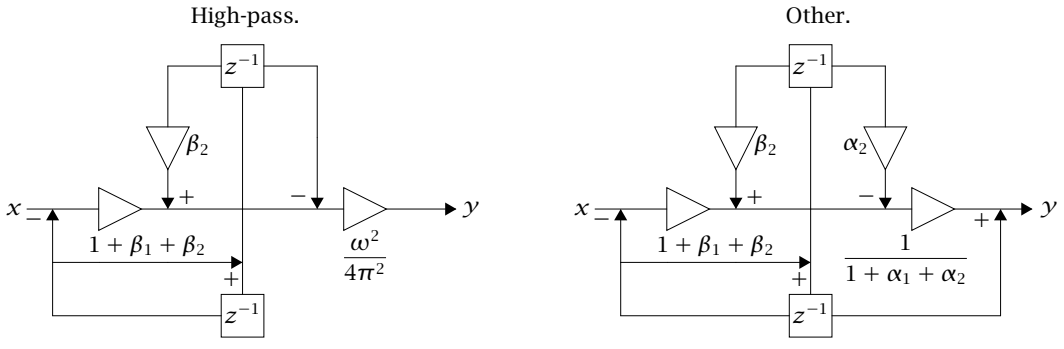


Figure 3.3. Optimised state space update for System 2.

We must, however, note that none of the systems from Table 3.3 are stable under time-varying conditions. Thus, some information is yet missing and it is to the question of stability that we now turn.

4. STABILITY

4.1. Definition and propositions

Let $\|\cdot\|_p$ denote the matrix norm induced by the vector norm $\|\cdot\|_p$ on $\mathbb{R}^m$ over $\mathbb{R}_+$.

DEFINITION 1. A system is said to be stable if every finite input produces a finite output.

$$\|\boldsymbol{x}_n\|_\infty < \infty \Longrightarrow \|\boldsymbol{y}_n\|_\infty < \infty \qquad \forall n \in \mathbb{R}$$

Let S denote a linear system defined as in (3.1), with $S \in \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}, \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}\}_n$ and $n \in \mathbb{N}$.

LEMMA 1. The output vector $\boldsymbol{y}_n$ is finite if the matrices $\boldsymbol{A}_n$ and $\boldsymbol{B}_n$, as well as the vector $\boldsymbol{z}_n$, are finite.

Proof.

$$\|\boldsymbol{y}_n\|_\infty = \|\boldsymbol{A}_n\boldsymbol{x}_n + \boldsymbol{B}_n\boldsymbol{z}_n\|_\infty \leqslant \underbrace{\|\boldsymbol{A}_n\|_\infty}_{<\infty}\underbrace{\|\boldsymbol{x}_n\|_\infty}_{<\infty} + \underbrace{\|\boldsymbol{B}_n\|_\infty}_{<\infty}\underbrace{\|\boldsymbol{z}_n\|_\infty}_{<\infty}$$ □

The condition for stability reduces to the behaviour of the state vector $\boldsymbol{z}_n$. Since $\boldsymbol{z}_{n+1}$ is defined recursively, it suffices to show that the sequence is defined for all n and that it does not diverge.

LEMMA 2. The sequence $\boldsymbol{z}_{n+1}$ is defined for all n if the matrices $\boldsymbol{C}_n$ and $\boldsymbol{D}_n$ are finite.

Proof.

$$\|\boldsymbol{z}_{n+1}\|_\infty = \|\boldsymbol{C}_n\boldsymbol{x}_n + \boldsymbol{D}_n\boldsymbol{z}_n\|_\infty \leqslant \underbrace{\|\boldsymbol{C}_n\|_\infty}_{<\infty}\underbrace{\|\boldsymbol{x}_n\|_\infty}_{<\infty} + \underbrace{\|\boldsymbol{D}_n\|_\infty}_{<\infty}\underbrace{\|\boldsymbol{z}_n\|_\infty}_{<\infty}$$ □

LEMMA 3. The sequence $\boldsymbol{z}_{n+1}$ converges if the product of the matrices $\boldsymbol{D}_0, \boldsymbol{D}_1, \ldots, \boldsymbol{D}_{n-1}, \boldsymbol{D}_n$ converges to the null matrix.

Proof. Expanding $\boldsymbol{z}_{n+1}$ and passing to the limit, one finds

$$\lim_{n\to\infty} \boldsymbol{z}_{n+1} = \lim_{n\to\infty}\left(\prod_{i=0}^{n} \boldsymbol{D}_i \cdot \boldsymbol{z}_0 + \sum_{i=0}^{n}\prod_{j=1}^{n-i} \boldsymbol{D}_j \cdot \boldsymbol{C}_i\boldsymbol{x}_i\right) = \prod_{i=1}^{\infty} \boldsymbol{D}_i \cdot \left(\boldsymbol{D}_0\boldsymbol{z}_0 + \sum_{i=0}^{\infty} \boldsymbol{C}_i\boldsymbol{x}_i\right)$$

which converges if there is absorption by $\mathbf{0}$. □

From Lemmata 1, 2 and 3 taken together, the following theorem is obtained.

THEOREM 1. A system S is stable if the matrices $\boldsymbol{A}_n$, $\boldsymbol{B}_n$ and $\boldsymbol{C}_n$ are finite and if the product of the matrices $\boldsymbol{D}_0, \boldsymbol{D}_1, \ldots, \boldsymbol{D}_{n-1}, \boldsymbol{D}_n$ converges to the null matrix.

Let us now assume that all state matrices of the system S are finite. We shall derive sufficient conditions for stability separately for the time-invariant and the time-variant cases.

PROPOSITION 1. A time-invariant system S is stable if the spectral radius of the matrix $\boldsymbol{D}$ is less than 1.

Proof. If $\rho(\boldsymbol{D}) < 1$ and $n \to \infty$, then $\rho(\boldsymbol{D}^n) = \rho(\boldsymbol{D})^n \to 0$. It follows that $\boldsymbol{D}^n$ is nilpotent, thus $\boldsymbol{D}^n \to \mathbf{0}$. □

For the time-variant case, however, greater caution is required. It is possible to construct a product $\boldsymbol{D}_0\boldsymbol{D}_1\cdots\boldsymbol{D}_{n-1}$ that diverges, but with $\boldsymbol{D}_n = 0$, mathematically, the system is still stable. Yet, in the real world, the energy accumulated inside a capacitor would break it, or the state vector $\boldsymbol{z}_n$ would overflow the available memory. Therefore, the best way to fight an enemy is to not have one.

PROPOSITION 2. A time-variant system S is stable if the spectral norm of the matrix $\boldsymbol{D}_n$ is less than 1.

Proof. Clearly, given any matrix norm $\|\cdot\|_p < 1$,

$$\prod \|\boldsymbol{D}_n\|_p \to 0 \implies \prod \boldsymbol{D}_n \to \boldsymbol{0}$$

Since $\|\boldsymbol{D}_n\|_2^2 = \rho\left(\boldsymbol{D}_n\boldsymbol{D}_n^{\mathsf{T}}\right) \leqslant \|\boldsymbol{D}_n\boldsymbol{D}_n^{\mathsf{T}}\|_p$, the lower bound to the matrix norm is attained for $p = 2$. □

Nevertheless, stable systems with a spectral norm equal to 1 do exist. This special case tells us that there exists a non-null vector $\boldsymbol{z}$, such that $\|\boldsymbol{D}_n\boldsymbol{z}\|_2 = \|\boldsymbol{z}\|_2$; that is, the transformation preserves the magnitude of the vector. The question is, however, whether $\|\boldsymbol{D}_n\boldsymbol{D}_{n+1}\cdots\boldsymbol{z}\|_2 < \|\boldsymbol{z}\|_2$. Rather than examining the matrix product sequentially, we settle the question by exhaustion through two examples.

EXAMPLE 1. Let there be given a real vector $\boldsymbol{z} = \begin{Vmatrix} z_1 \\ z_2 \end{Vmatrix}$ with $z_1 \neq 0$ and $z_2 \neq 0$. Then the only magnitude-preserving linear transformation of such a vector is a rotation or a reflection, which has determinant equal to ± 1.

EXAMPLE 2. Let us take the same vector $\boldsymbol{z}$, but with $z_2 = 0$ instead. We have a case where the 2-dimensional vector $\boldsymbol{z}$ lies in the span of the set $P_2 = \left\{\begin{Vmatrix} 1 \\ 0 \end{Vmatrix}, \begin{Vmatrix} 0 \\ 1 \end{Vmatrix}\right\}$, but also in the span of the set $P_1 = \left\{\begin{Vmatrix} 1 \\ 0 \end{Vmatrix}\right\}$. If rotation is excluded, then a magnitude-preserving transformation on the vector $\boldsymbol{z}$ is possible with a shift. If the order of the shift is reversed by a cycle, oscillation may occur. For instance, let us choose

$$\boldsymbol{D_1} = \begin{Vmatrix} 0 & 0 \\ 1 & 0 \end{Vmatrix} \quad \text{and} \quad \boldsymbol{D_2} = \begin{Vmatrix} 0 & 1 \\ 0 & 0 \end{Vmatrix}$$

The entries of the vector $\boldsymbol{z}$ are thereby permutable cyclically:

$$\boldsymbol{D_1 z} = \begin{Vmatrix} 0 \\ z_1 \end{Vmatrix}, \quad \boldsymbol{D_2 D_1 z} = \begin{Vmatrix} z_1 \\ 0 \end{Vmatrix}, \quad \boldsymbol{D_1 D_2 D_1 z} = \begin{Vmatrix} 0 \\ z_1 \end{Vmatrix}, \quad \ldots$$

Instead of trying to identify a cyclic shift, it is simpler to not allow it in the first place. In fact, that combinatorial phenomenon is entirely absent in analogue filters. Therefore, a sufficient condition for its non-occurrence is that no off-diagonal entry of the matrix equal ± 1.

From exhaustion, the final proposition is obtained.

PROPOSITION 3. The following holds:

$$\left.\begin{aligned} &\|\boldsymbol{D}_n\|_2 = 1 \\ &\det(\boldsymbol{D}_n) \neq \pm 1 \\ &\max|\boldsymbol{D}_{n|ij}| \neq 1 \end{aligned}\right\} \implies \prod \boldsymbol{D}_n \to \boldsymbol{0}$$

4.2. The state variable form

There exists an infinite number of stable time-variant second-order structures, as the preceding section makes clear. Therefore, in order to start from somewhere, it is natural to take as a starting point a simple continuous form whose stability is already assured. One of such systems is characterised by the following continuous transition matrix $\boldsymbol{D}_t$ and equilibrium matrix $\boldsymbol{E}$:

$$\boldsymbol{D}_t = \begin{Vmatrix} 0 & 1 \\ -1 & -2\zeta \end{Vmatrix} \cdot 2\pi f_c, \qquad \boldsymbol{E} = \begin{Vmatrix} 1 \\ 0 \end{Vmatrix} \tag{4.1}$$

The literature calls this structure State Variable Form. We recall that the convergence of the time-variant transition matrix depends not only on its eigenvalues, but also on its eigenvectors. To discretise $\boldsymbol{D}_t$, we simply eigendecompose it, transform its eigenvalues, and recompose:

$$\boldsymbol{D}_t = \boldsymbol{V}_t \boldsymbol{\Lambda}_t \boldsymbol{V}_t^{-1} \quad \Longrightarrow \quad \boldsymbol{D}_n = \boldsymbol{V}_t \boldsymbol{\Lambda}_n \boldsymbol{V}_t^{-1} \tag{4.2}$$

In fact, with the continuous eigenvector matrix $\boldsymbol{V}_t$ and discrete eigenvalue matrix $\boldsymbol{\Lambda}_n$

$$\boldsymbol{V}_t = \begin{Vmatrix} 1 & 1 \\ -\zeta + \sqrt{\zeta^2 - 1} & -\zeta - \sqrt{\zeta^2 - 1} \end{Vmatrix}, \qquad \boldsymbol{\Lambda}_n = \frac{1}{2} \cdot \begin{Vmatrix} -\beta_1 + \sqrt{\beta_1^2 - 4\beta_2} & 0 \\ 0 & -\beta_1 - \sqrt{\beta_1^2 - 4\beta_2} \end{Vmatrix} \tag{4.3}$$

the discrete state space is then constructed step by step:

$$\boldsymbol{D}_n = \boldsymbol{V}_t \boldsymbol{\Lambda}_n \boldsymbol{V}_t^{-1} \quad \longrightarrow \quad \boldsymbol{C}_n = (\boldsymbol{I} - \boldsymbol{D}_n)\boldsymbol{E} \quad \longrightarrow \quad \boldsymbol{B}_n = \begin{Vmatrix} (1 + \beta_1 + \beta_2)(\alpha_1 - \beta_1) \\ \alpha_1 + \alpha_2 - \beta_1 - \beta_2 \end{Vmatrix}^{\mathsf{T}} \cdot \begin{Vmatrix} \boldsymbol{C}_n & \boldsymbol{E} \end{Vmatrix}^{-1} \tag{4.4}$$

Alternatively, $\boldsymbol{V}_t$ may be expressed in terms of the elements of the set $\{1, \beta_1, \beta_2\}$. Depending on the coefficient mapping, this might be tedious. Nevertheless, there is no requirement to exactly match $\boldsymbol{V}_t$, as shown in Figure 5.1. For instance, setting $\sigma = 0$, one obtains

$$\zeta = \sqrt{\frac{\nu + \kappa}{2\nu}} \tag{4.5}$$

Then, with $\mu = \sqrt{\nu}/\pi$, one arrives at the following elegant state space:

$$\begin{cases} \boldsymbol{A} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \\ \boldsymbol{B} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \cdot \begin{Vmatrix} -1 \\ -\mu \end{Vmatrix}^{\mathsf{T}} + \begin{Vmatrix} 1 + \alpha_1 + \alpha_2 \\ 1 - \alpha_1 + \alpha_2 \end{Vmatrix}^{\mathsf{T}} \cdot \begin{Vmatrix} 1 & 0 \\ 0 & \mu^{-1} \end{Vmatrix} \\ \boldsymbol{C} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \cdot \begin{Vmatrix} 1 \\ -\mu \end{Vmatrix} \cdot \frac{1}{2} \\ \boldsymbol{D} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \cdot \begin{Vmatrix} 1 \\ -\mu \end{Vmatrix} \cdot \frac{1}{2} \cdot \begin{Vmatrix} -1 \\ -\mu \end{Vmatrix}^{\mathsf{T}} + \begin{Vmatrix} 1 & 0 \\ 0 & -1 \end{Vmatrix} \end{cases} \tag{4.6}$$

Furthermore, with the auxiliary quantity

$$\vartheta = (1 + \beta_1 + \beta_2)(x - \boldsymbol{z}_{1|n} - \mu \boldsymbol{z}_{2|n}) \tag{4.7}$$

the system updates as shown in Table 4.1:

Table 4.1.
Optimised state space update for the state variable form.

	$\boldsymbol{y}_n$	$\boldsymbol{z}_{1\|n+1}$	$\boldsymbol{z}_{2\|n+1}$
Low Cut	$G \cdot \left(\vartheta + \frac{1 - \alpha_1 + \alpha_2}{\mu} \cdot \boldsymbol{z}_{2\|n} \right)$	$\boldsymbol{z}_{1\|n} + \frac{\vartheta}{2}$	$-\boldsymbol{z}_{2\|n} - \mu \cdot \frac{\vartheta}{2}$
Other	$G \cdot \left(\vartheta + \frac{1 - \alpha_1 + \alpha_2}{\mu} \cdot \boldsymbol{z}_{2\|n} \right) + \boldsymbol{z}_{1\|n}$	$\boldsymbol{z}_{1\|n} + \frac{\vartheta}{2}$	$-\boldsymbol{z}_{2\|n} - \mu \cdot \frac{\vartheta}{2}$

4.3. The eigenmorph form

A direct algebraic operation of the eigenvector matrix $\boldsymbol{V}_t$ and the equilibrium matrix $\boldsymbol{E}$ in (4.1), with $\eta \in \mathbb{R}$, yields

$$\boldsymbol{V}_t = \begin{Vmatrix} 1 & 1 \\ -\zeta + \sqrt{\zeta^2 - 1} - \eta & -\zeta - \sqrt{\zeta^2 - 1} - \eta \end{Vmatrix} \quad \text{and} \quad \boldsymbol{E} = \begin{Vmatrix} 1 \\ -\eta \end{Vmatrix} \tag{4.8}$$

This form is then stable for time-varying coefficients when $\eta \in \left[0, 2\sqrt{\frac{\nu+\kappa}{2\nu}}\right] \cong [0, 2\zeta]$. The interesting part, however, is that it nicely extends the state space of the state variable form in (4.6). At the expense of three more additions and one multiplication, we have a modular structure that maintains its frequency response.

$$\begin{cases} \boldsymbol{A} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \\ \boldsymbol{B} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \cdot \begin{Vmatrix} -\eta\mu - 1 \\ -\mu \end{Vmatrix}^{\mathsf{T}} + \begin{Vmatrix} 1 + \alpha_1 + \alpha_2 \\ 1 - \alpha_1 + \alpha_2 \end{Vmatrix}^{\mathsf{T}} \cdot \begin{Vmatrix} 1 & \eta\mu^{-1} \\ 0 & \mu^{-1} \end{Vmatrix} \\ \boldsymbol{C} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \cdot \begin{Vmatrix} 1 \\ -\eta - \mu \end{Vmatrix} \cdot \frac{1}{2} \\ \boldsymbol{D} = \begin{Vmatrix} 1 + \beta_1 + \beta_2 \end{Vmatrix} \cdot \begin{Vmatrix} 1 \\ -\eta - \mu \end{Vmatrix} \cdot \frac{1}{2} \cdot \begin{Vmatrix} -\eta\mu - 1 \\ -\mu \end{Vmatrix}^{\mathsf{T}} + \begin{Vmatrix} 1 & 0 \\ -2\eta & -1 \end{Vmatrix} \end{cases} \tag{4.9}$$

Furthermore, with the auxiliary quantity

$$\vartheta = (1 + \beta_1 + \beta_2)(x - \boldsymbol{z}_1 - \mu(\eta \boldsymbol{z}_{1|n} + \boldsymbol{z}_{2|n})) \tag{4.10}$$

the system is updated as shown in Table 4.2:

Table 4.2.
Optimised state space update for the eigenmorph form.

	$\boldsymbol{y}_n$	$\boldsymbol{z}_{1\|n+1}$	$\boldsymbol{z}_{2\|n+1}$
Low Cut	$G \cdot \left(\vartheta + \frac{(1-\alpha_1+\alpha_2)(\eta \boldsymbol{z}_{1\|n} + \boldsymbol{z}_{2\|n})}{\mu} \right)$	$\boldsymbol{z}_{1\|n} + \frac{\vartheta}{2}$	$-\boldsymbol{z}_{2\|n} - (\eta+\mu) \cdot \frac{\vartheta}{2} - 2\eta \boldsymbol{z}_{1\|n}$
Other	$G \cdot \left(\vartheta + \frac{(1-\alpha_1+\alpha_2)(\eta \boldsymbol{z}_{1\|n} + \boldsymbol{z}_{2\|n})}{\mu} \right) + \boldsymbol{z}_{1\|n}$	$\boldsymbol{z}_{1\|n} + \frac{\vartheta}{2}$	$-\boldsymbol{z}_{2\|n} - (\eta+\mu) \cdot \frac{\vartheta}{2} - 2\eta \boldsymbol{z}_{1\|n}$

The structural symmetry of Figure 4.1 and Figure 4.2 is worth remarking upon; both obey the same natural pattern, and certain gain blocks admit further simplification, for it can be shown that

$$1 + \beta_1 + \beta_2 = \frac{4\pi^2}{\pi^2 + \pi\sqrt{2}\sqrt{\nu+\kappa} + \nu} \tag{4.11}$$

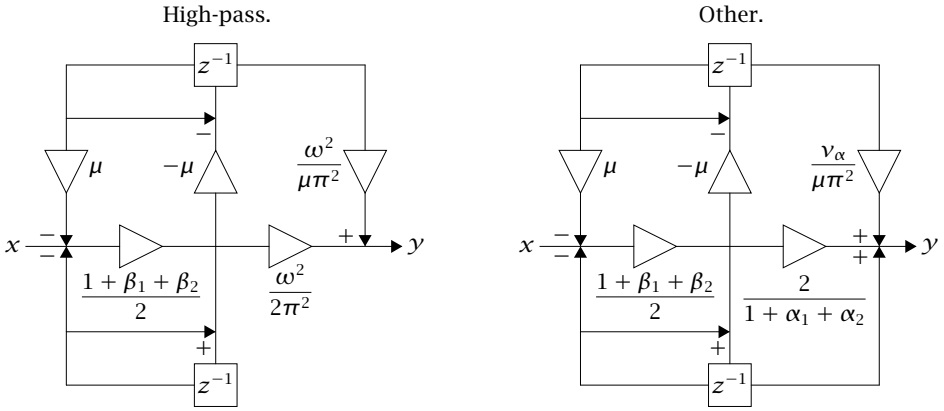


Figure 4.1. Optimised state variable form.

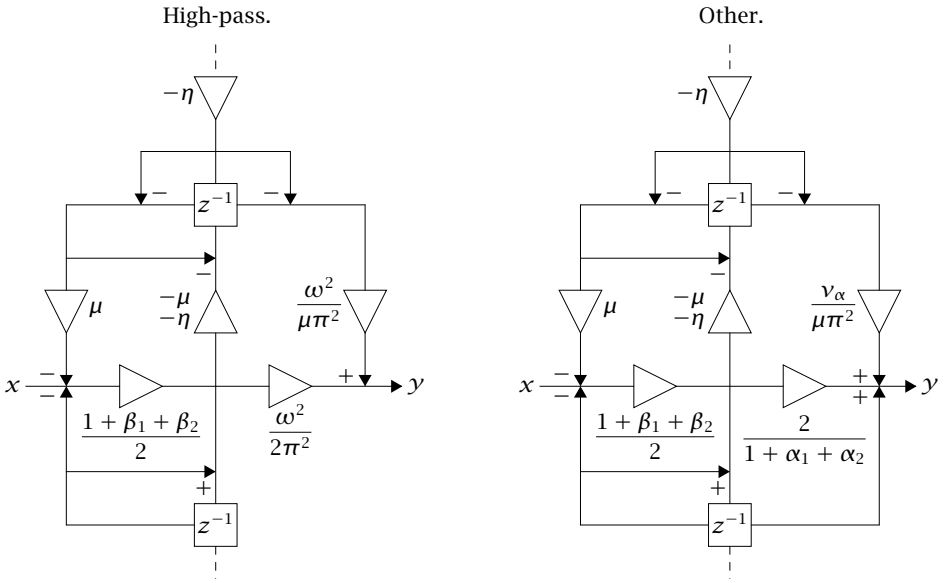


Figure 4.2. Optimised eigenmorph form.

4.4. The normal form

We conclude with a form of particular theoretical interest. Normal matrices are of great utility in mathematics by virtue of their symmetric properties; in particular, they possess an orthonormal basis of eigenvectors. For a second-order system, this means that the two states are exactly 90 degrees out of phase, a property that guarantees stability and stands in instructive contrast to the Direct Form, whose states are phase-shifted by one sample, as Figure 5.2 shows.

In fact, with the eigenvector and equilibrium matrices

$$\boldsymbol{V} = \begin{Vmatrix} 1 & 0 \\ 0 & 1 \end{Vmatrix} \quad \text{and} \quad \boldsymbol{E} = \begin{Vmatrix} 1 \\ 1 \end{Vmatrix} \tag{4.12}$$

the state space can directly be built from the discrete eigenvalues λ_1 and λ_2:

$$\begin{cases} \boldsymbol{A} = \begin{Vmatrix} (\lambda_1 - 1)(\lambda_2 - 1) \end{Vmatrix} \\ \boldsymbol{B} = \begin{Vmatrix} (\lambda_1^2 + \lambda_1\alpha_1 + \alpha_2)(1 - \lambda_2) \\ (\lambda_2^2 + \lambda_2\alpha_1 + \alpha_2)(\lambda_1 - 1) \end{Vmatrix}^{\mathsf{T}} \cdot \dfrac{1}{\lambda_1 - \lambda_2} \\ \boldsymbol{C} = \begin{Vmatrix} 1 - \lambda_1 \\ 1 - \lambda_2 \end{Vmatrix} \\ \boldsymbol{D} = \begin{Vmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{Vmatrix} \end{cases} \tag{4.13}$$

This form excludes the critically damped case $\zeta = 1$, for which $\lambda_1 = \lambda_2$ and where a Jordan form is required.

REFERENCE

[1] Yuriy Ivantsov, “On the ideal bilinear and biquadratic digital filter.” [Online]. Available: https://ivantsovy.com/research/paper1.pdf

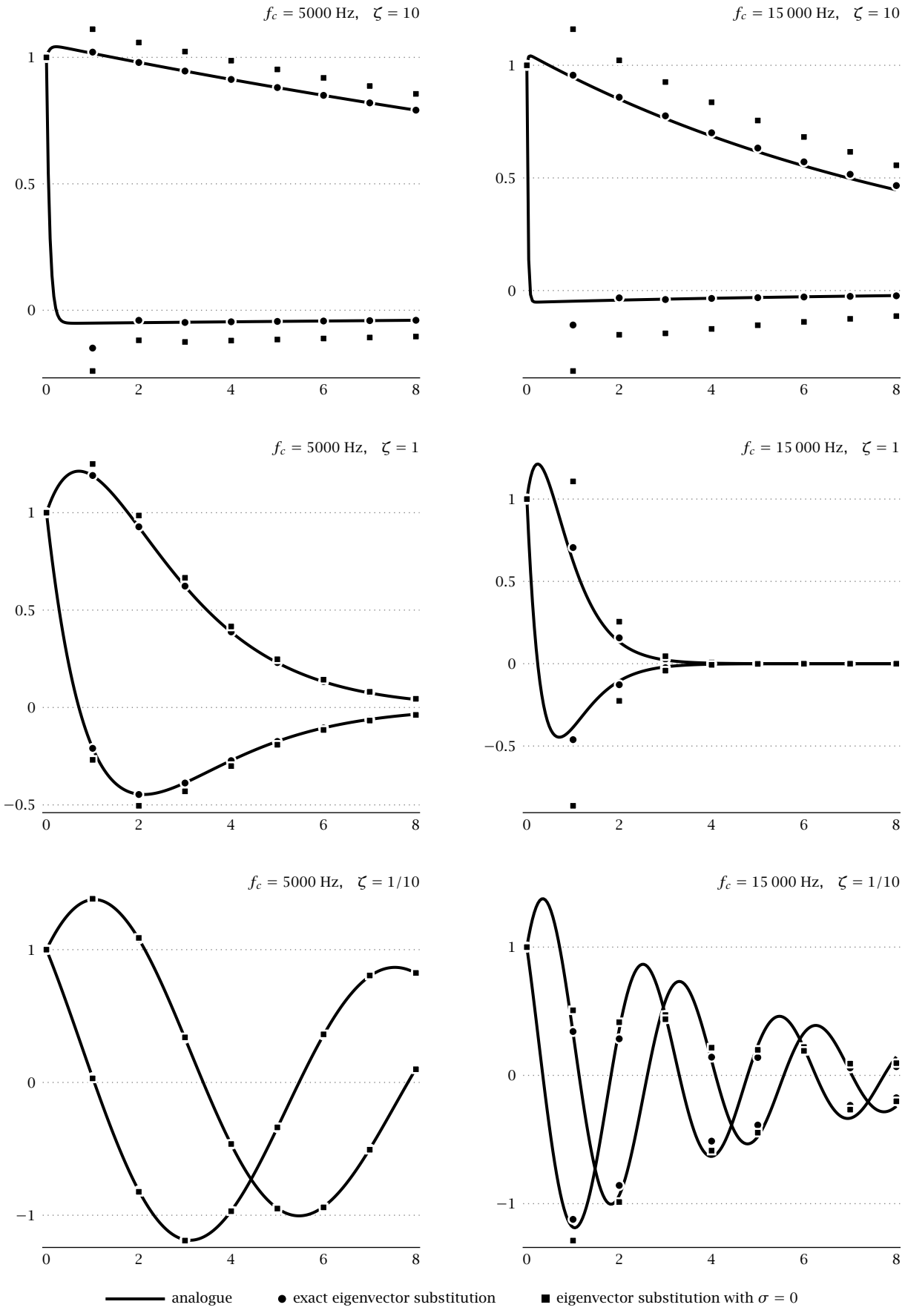


Figure 5.1. Comparison of evolution of states in the state variable form, based on the eigenvector discretisation.
$f_s = 44.1$ kHz, $\sigma = \pi\sqrt{2/3}$, $\boldsymbol{z}_0 = \|1 \quad 1\|^\mathsf{T}$

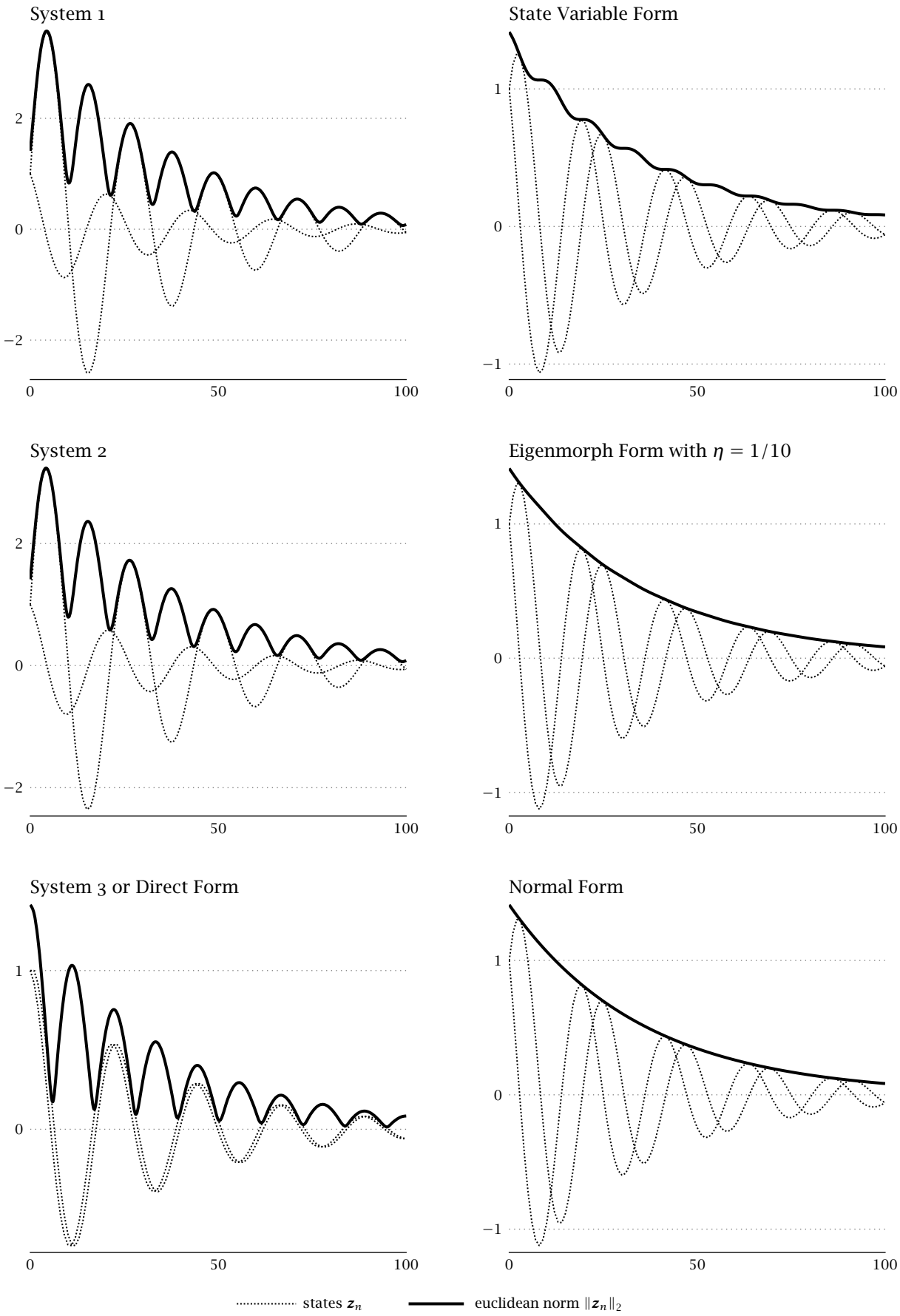


Figure 5.2. Evolution of states across various structures. On the left, a source of potential instability: humps.
$f_c = 2000$ Hz, $f_s = 44.1$ kHz, $\zeta = 1/10$, $\sigma = \pi\sqrt{2/3}$, $z_0 = \|1 \quad 1\|^\mathsf{T}$